MODULE 4A · EPISODE 2

Kepler's Laws and Their Applications

From Tycho's Mars Data to Weighing the Stars

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Kepler's Laws and Their Applications

This episode turns to Johannes Kepler, the man who made the heliocentric model work. Using Tycho Brahe's careful observations of Mars, Kepler found three laws of planetary motion: orbits are ellipses with the Sun at one focus, a planet sweeps out equal areas in equal times, and the square of a planet's year is proportional to the cube of its average distance from the Sun. The panel shows how the Second Law shapes Earth's own seasons, and how the Third Law, combined with the transits of Venus, revealed the size of the entire Solar System. It then puts gravity to work in a pair of thought experiments, follows the Moon from its violent birth to its present distance and its locked rotation, and closes with the Third Law's most powerful modern use: measuring the masses of stars in binary systems many light-years away.

The Panel

Name	Role
Dr. Edward Thorne	Host; astrophysicist
Dr. Lena Petrova	Guest; exoplanet researcher
Dr. Kenji Tanaka	Guest; microbial ecologist

Learning Objectives

After this episode you should be able to:

- Explain why Kepler holds a unique place in astronomy, and describe the role of Tycho Brahe's observations.
- State Kepler's First Law and explain how it corrected Copernicus's model.
- State Kepler's Second Law and explain why a planet moves fastest when it is closest to the Sun.
- Describe how the Second Law affects Earth's daylight and the length of its seasons.
- State Kepler's Third Law and explain the meaning of orbital period and semi-major axis.
- Describe how Newton's laws provided the theoretical foundation for Kepler's Laws.
- Explain how the transits of Venus measured the astronomical unit, and how the Third Law then gave the size of the Solar System.
- Use Newton's gravity and his Third Law to explain why everyday gravitational forces are tiny and why the Earth doesn't rise to meet you when you jump.
- Describe the Moon's formation, its outward migration, the lengthening of Earth's day, and tidal locking.
- Explain how Kepler's Third Law, eclipses, and Doppler shifts reveal the masses of binary stars.

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Key Terms

Term	Definition
Kepler's First Law	The orbit of a planet is an ellipse, with the Sun at one of the ellipse's two foci.
Ellipse	An oval curve, a “squished circle,” with two foci.
Focus (plural: foci)	One of the two fixed points inside an ellipse. The Sun sits at one focus of a planet's orbit; the other focus is empty.
Kepler's Second Law	A line joining a planet and the Sun sweeps out equal areas during equal intervals of time.
Kepler's Third Law	The square of a planet's orbital period is directly proportional to the cube of the semi-major axis of its orbit.
Orbital period	The time a planet takes to complete one orbit; the planet's year.
Semi-major axis	Half the longest diameter of an ellipse; for a planet, its average distance from the Sun.
Solstice	One of the two times each year, in June and December, when the Sun is farthest north or south in the sky.
Transit of Venus	A rare event in which Venus passes across the face of the Sun as seen from Earth.
Astronomical unit (au)	The average distance between the Earth and the Sun, about 93 million miles.
Newton's Law of Universal Gravitation	Every two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between their centers.
Newton's Third Law of Motion	For every action, there is an equal and opposite reaction.
Tidal forces	Forces that come from the difference in gravity's pull across a body. They pushed the Moon outward and slowed its rotation.
Angular momentum	The “rotational momentum” of a spinning or orbiting body. Earth has transferred some of its spin angular momentum to the Moon.
Tidal locking	The state in which a body rotates on its axis exactly once for every revolution it makes around its partner, so it always shows the same face.
Near side	The side of the Moon that always faces Earth.
Binary star system	Two stars orbiting each other. Such systems are very common.
Eclipsing binary	A binary star system in which the stars pass in front of each other as seen from Earth, causing dips in the system's total brightness.
Doppler shift	The change in the wavelength of light from a moving source: shorter (bluer) when the source moves toward us, longer (redder) when it moves away.

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Term	Definition
Light-year	The distance light travels in one year, about 6 trillion miles.

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Numbers to Know

Value	What it refers to
Early January	When Earth is closest to the Sun, and moving fastest in its orbit
~4 days	How much shorter Northern Hemisphere winter is than summer, the longest season
1761, 1769	The transits of Venus used to measure the au
~95 million mi	Estimate of the au from the combined transit observations
93 million mi	Modern value of the au
80 kg, 1,589 kg, 1 m	The person and car in the gravity thought experiment
8.5×10^{-6} N	Their gravitational attraction, about the weight of a small worker ant
1.3×10^{-22} m/s²	The acceleration an 80 kg person gives the Earth
~4 billionths	How far the Earth rises, as a fraction of a hydrogen nucleus's diameter, when you jump half a meter
20,000–30,000 km	The Moon's distance from Earth when it formed
384,000 km	The Moon's average distance today
4.5 billion years	The time over which tidal forces moved the Moon outward
4–6 h → 24 h	The length of Earth's day then and now
~1 Earth month	The length of one Moon day

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EPISODE CONTENT

Part-by-Part Summary

Part 1 — Kepler's Laws of Planetary Motion

Johannes Kepler holds a unique and revered place in the history of astronomy. He paved the way for the acceptance of the heliocentric model by “fixing” the Copernican model of the Solar System, and his laws came not from theory, but from careful observational data.

A contemporary of Galileo, Kepler worked for **Tycho Brahe**, the last great naked-eye astronomer. When Brahe died, Kepler inherited—some would say took—his priceless observations of Mars, which allowed him to work out his three laws of planetary motion.

Kepler's First Law states that the orbit of a planet is an **ellipse**, with the Sun at one of the ellipse's two **foci**. Think of an orbit as a squished circle. This was the key insight that corrected Copernicus's erroneous assumption of perfectly circular orbits.

Part 2 — The Second and Third Laws

Kepler's Second Law states that a line segment joining a planet and the Sun sweeps out **equal areas during equal intervals of time**. So when a planet is closer to the Sun, it must move faster.

This affects us directly. Earth is closest to the Sun in **early January**, so it moves fastest around the December solstice. That is why the day-to-day changes in daylight around the December solstice are slightly larger than those around the June solstice, and why Northern Hemisphere **winter is the shortest season**, by about four days.

Kepler's Third Law, perhaps the most valuable to astronomers, states that the square of a planet's **orbital period** is directly proportional to the cube of the **semi-major axis** of its orbit. In simpler terms, the square of a planet's year is proportional to the cube of its average distance from the Sun, a precise mathematical link between a planet's period and its distance.

Newton was later able to **derive all three of Kepler's Laws** from his own laws of motion and his law of gravity, providing the theoretical foundation that Kepler's work was missing.

Part 3 — Measuring the Solar System

Newton's friend **Edmond Halley** proposed that the **transits of Venus** in 1761 and 1769—rare events in which Venus passes across the face of the Sun—could be used to find the average distance between the Sun and the Earth, the unit we now call the **astronomical unit**, or **au**. Sadly, Halley didn't live to see either transit.

Numerous groups were sent to different locations on Earth to observe the transits. Their combined results gave an estimate of about **95 million miles**, remarkably close to the modern value of **93 million miles**.

Once the au was known, astronomers could use **Kepler's Third Law** to calculate the average distance to all the other planets, and so determine the size of the entire Solar System.

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Part 4 — Gravity in Action

How attractive are you, gravitationally speaking? The gravitational attraction between an **80-kilogram person** and a **1,589-kilogram car** standing one meter apart is about **8.5×10^{-6} newtons**, roughly the weight of a small worker ant. It's a tiny force.

When you jump, the Earth pulls you down, but by **Newton's Third Law** you also pull the Earth up with an equal force. The Earth doesn't come up to meet you because, although the forces are equal, the **accelerations are not**: the Earth's mass is enormous compared with yours. An 80-kilogram person gives the Earth an acceleration of only about **1.3×10^{-22} m/s²**. Jump half a meter high, and the Earth rises to meet you by only about **four billionths of the diameter of a hydrogen nucleus**.

Part 5 — The Earth-Moon System

The Moon formed from debris thrown off when a **Mars-sized object struck the early Earth**. At first it was only about **20,000 to 30,000 kilometers** away. Over 4.5 billion years, **tidal forces** have moved it outward to its current average distance of about **384,000 kilometers**.

The same process transferred some of Earth's rotational momentum, its **angular momentum**, to the Moon, lengthening our day from about **4 to 6 hours** to the **24 hours** we have now.

Early in its history, Earth's gravity slowed the Moon's rotation until it turned on its axis exactly once for every revolution around the Earth. This is **tidal locking**. As a result, one Moon day is about one Earth month long, and the same side, the **near side**, always faces us.

Part 6 — Kepler's Third Law in Modern Astronomy

Kepler's Third Law applies to **any two objects orbiting each other**, including **binary star systems**, which are very common. If we can measure both the period of the orbit and its size, the law gives the two stars' **combined mass**.

Eclipsing binaries are especially useful. When one star passes in front of the other, the system's total brightness dims, and timing those dips gives the **orbital period**. Combine that with the stars' orbital speeds, measured from **Doppler shifts** in their light, and we can work out the size of the orbit and even the **mass of each individual star**, a powerful tool for understanding stars many light-years away.

From elliptical orbits to the masses of distant stars, Kepler's Laws remain a cornerstone of astronomy.

TRY IT YOURSELF

Draw an ellipse. Push two pins into a sheet of cardboard a few inches apart, drop a loop of string over them, and pull it tight with a pencil. Keeping the string taut, trace all the way around. Each pin is a focus; put the Sun at one of them. Move the pins farther apart and draw again: the ellipse gets more squished. Push them together and it becomes a circle.

Then try the Third Law: Jupiter is about 5.2 au from the Sun. With $T^2 = a^3$, how long is Jupiter's year? (Answer: $5.2^3 \approx 140.6$, and $\sqrt{140.6} \approx 11.9$ years. The measured value is 11.86 years.)

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STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Kepler's Laws of Planetary Motion

1. Why does Kepler hold a unique place in the history of astronomy? Where did his laws come from?

2. Who was Tycho Brahe, and what did Kepler gain from him?

3. State Kepler's First Law of Planetary Motion.

4. What is an ellipse? Where is the Sun located in a planet's orbit, and what is at the other focus?

5. Which assumption in Copernicus's model did Kepler's First Law correct?

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Part 2 — The Second and Third Laws

1. State Kepler's Second Law of Planetary Motion.

2. What does the Second Law tell us about a planet's speed? Where in its orbit is a planet fastest?

3. When is Earth closest to the Sun, and how does this affect Earth's motion around the December solstice?

4. Describe two effects on Earth of its faster orbital motion around the December solstice.

5. State Kepler's Third Law, first formally and then in simpler terms.

6. When a planet's year is measured in Earth years and its distance in au, the Third Law becomes $T^2 = a^3$. A planet's average distance from the Sun is 4 au. How long is its year?

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7. What did Newton add to Kepler's Laws?

Part 3 — Measuring the Solar System

1. Who proposed using the transits of Venus to measure the Solar System, and what were they meant to determine?

2. What is a transit of Venus? In which years did the transits Halley proposed occur, and did he see them?

3. How were the transits observed, and how did the result compare with the modern value?

4. What is the astronomical unit?

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5. Once the au was known, how did astronomers determine the size of the entire Solar System?

Part 4 — Gravity in Action

1. How large is the gravitational force between an 80-kilogram person and a 1,589-kilogram car standing one meter apart? What is it comparable to?

2. Why is the gravitational attraction between everyday objects, like a person and a car, so tiny?

3. When you jump, the Earth pulls you down. According to Newton's Third Law, what do you do to the Earth?

4. If the forces are equal, why doesn't the Earth come up to meet you?

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5. If you jump half a meter high, about how far does the Earth rise to meet you?

Part 5 — The Earth-Moon System

1. How did the Moon form?

2. How far from Earth was the Moon when it formed, and how far is it now? What caused the change?

3. What did Earth transfer to the Moon as the Moon moved outward, and how did this change the length of Earth's day?

4. What is tidal locking, and how did it happen to the Moon?

5. Why do we always see the same side of the Moon? How long is one day on the Moon?

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Part 6 — Kepler's Third Law in Modern Astronomy

1. To which systems besides planets orbiting the Sun can Kepler's Third Law be applied? Give an example.

2. Which two measurements of a binary star's orbit let us find the stars' combined mass?

3. What is an eclipsing binary, and how does it reveal its orbital period?

4. How are the stars' orbital speeds measured, and what do they add to the period?

5. The episode calls Kepler's Laws “a cornerstone of astronomy.” Use examples from the episode to explain why.

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YOUR NOTES

Notes

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